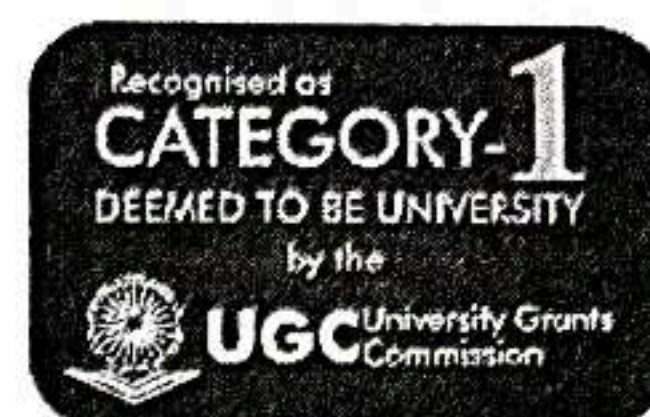
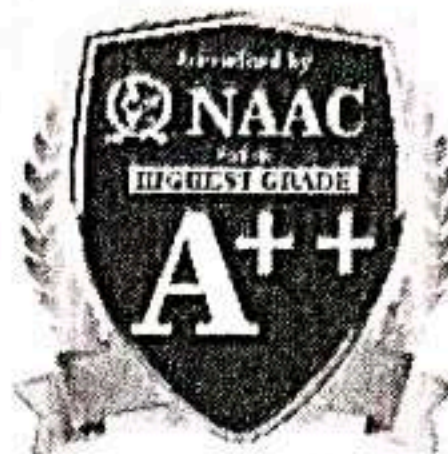




Vel Tech
Rangarajan Dr. Sagunthala
R&D Institute of Science and Technology
(Deemed to be University Estd. u/s 3 of UGC Act, 1956)



School of Computing
Department of CSE (Data Science)
ACADEMIC YEAR 2025 – 2026 (Summer Semester)

BONAFIDE CERTIFICATE

NAME: P. Koteswara Rao

VTU NO: VTU28129

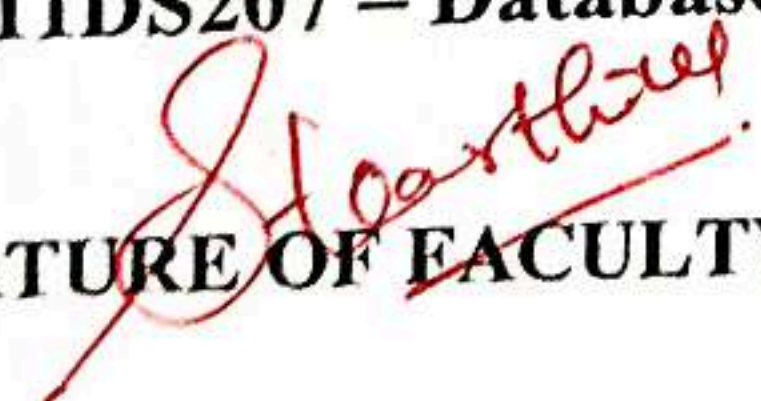
REG.NO: 24UEDLOO53

BRANCH: B.Tech CSE (DATA SCIENCE)

YEAR/SEM: 2nd year / 3rd sem

SLOT: S15-12

Certified that this is a bonafide record of work done by above student in the
"10211DS207 – Database Management Systems" during the year 2025-2026.


SIGNATURE OF FACULTY INCHARGE

SIGNATURE OF INCHARGE

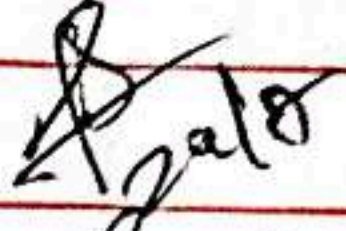
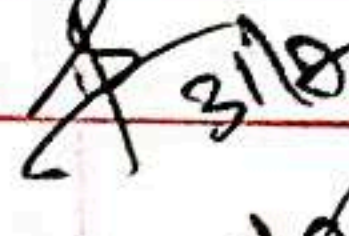
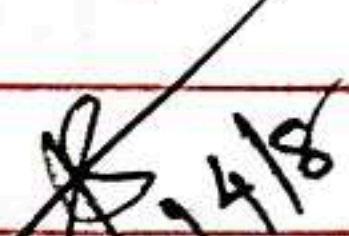
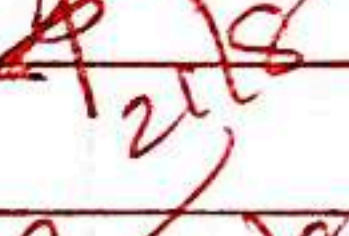
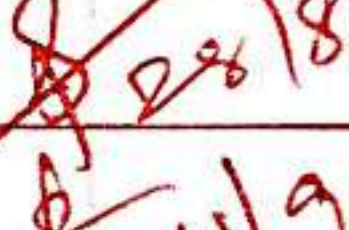
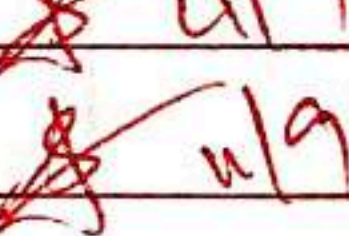
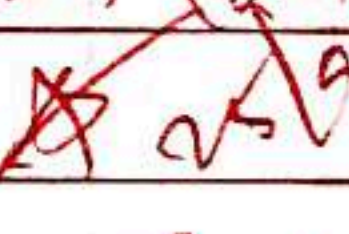
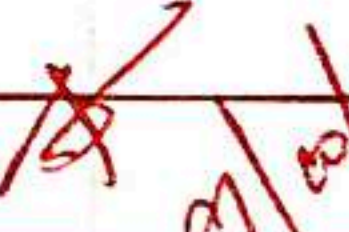
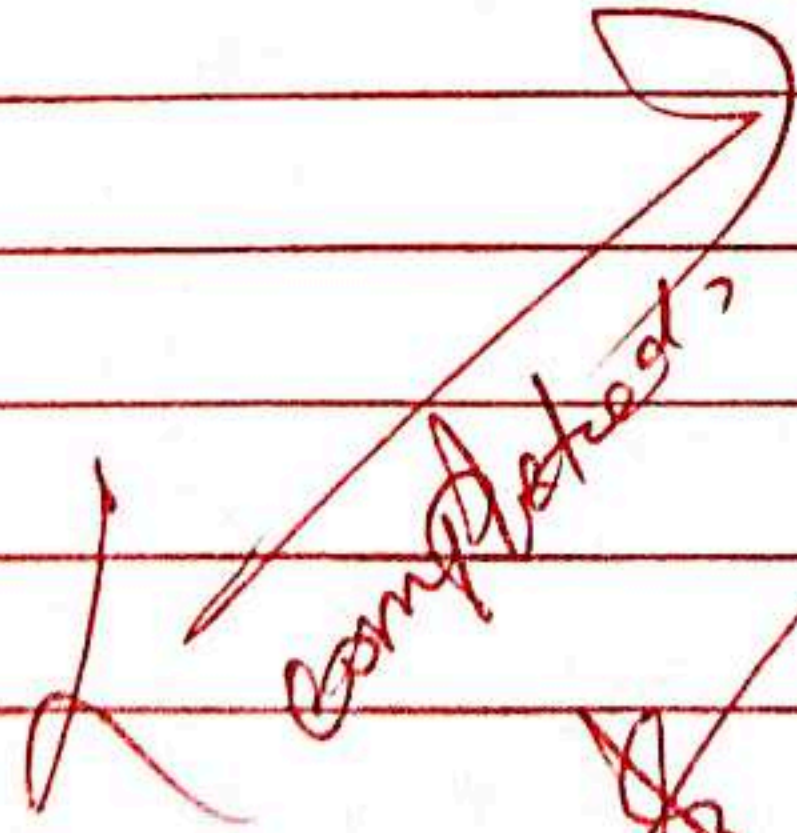
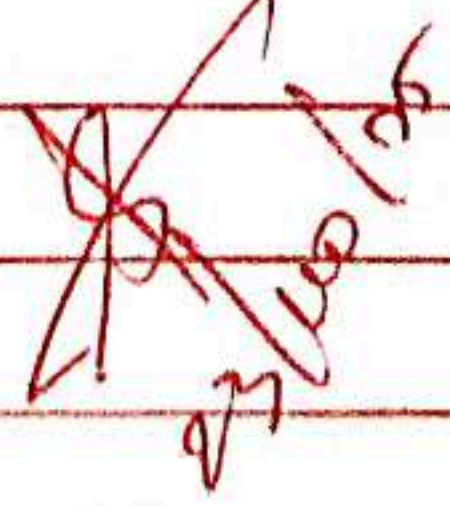
Submitted for the Semester Model Examination held on _____ at Vel Tech
Rangarajan Dr. Sagunthala R & D Institute of Science and Technology.

EXAMINER 1

EXAMINER 2

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Name: P. Kotegowara Rao Class: IInd year. Subject DBMS LAB

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4.	04/08/2025	Developing Queries with DML Multi-Row Functions and operations	20	
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TASK-1

31/07/2025

A Bank Management system is a comprehensive software solution designed to manage and streamline banking operations. It covers various aspects of banking, including customer account management, transaction processing, loan and mortgage management and more.

AIM:- To draw Conceptual Design through PTR using drawio tool.

Procedure:-

Step 1(a): Identifying the Entities

Entities represent the major objects in the banking system.

Sample output:-

- * customer
- * Account
- * Transaction
- * Loan
- * Branch
- * Employee

Step 1(b): Identifying the Attributes

For each entity, list out the attributes

(primary key is underlined)

Sample output:-

1. customer

→ customer ID (PK)

→ Name

→ Address

→ phone

→ email

2. Account

→ Account - Number (PK)

→ Account - Type

→ Balance

→ Date - opened

3. Transaction

→ Transaction - ID (PK)

→ Transaction - Type

→ Amount

→ Date - Time

4. Loan

→ Loan - ID (PK)

→ Loan - Type

→ Loan - Amount

→ Interest - Rate

5. Branch

→ Branch - ID (PK)

→ Branch - Name

→ Location

6. Employee

→ Employee - ID (PK)

→ Name

→ Position

→ Contact - No

Step 1 (c): Identifying Relationships, Cardinality and Type

1. Customer - Account

* A customer has one or more accounts

* Cardinality: One to Many.

* Type: strong relationship

2. Account - Transaction

* An account performs many transactions

* cardinality: One to Many (1:M)

* Type: strong relationship

2. Account

→ Account - Number (PK)

→ Account - Type

→ Balance

→ Date - opened

3. Transaction

→ Transaction - ID (PK)

→ Transaction - Type

→ Amount

→ Date - Time

4. Loan

→ Loan - ID (PK)

→ Loan - Type

→ Loan - Amount

→ Interest - Rate

5. Branch

→ Branch - ID (PK)

→ Branch - Name

→ Location

6. Employee

→ Employee - ID (PK)

→ Name

→ Position

→ Contact - No

Step 1 (c): Identifying Relationships, Cardinality and Type

1. Customer - Account

* A customer has one or more accounts

* Cardinality: One to Many

* Type: strong relationship

2. Account - Transaction

* An account performs many transactions

* cardinality: One to Many (1:M)

* Type: strong relationship

3. customer - loan

* A customer takes zero or many loans

* cardinality : One to Many (1:M)

* Type : strong relationship

4. Branch - employee

* A Branch employs many employees

* Cardinality : One to Many

5. Branch - Account

* A Branch manages many accounts

* Cardinality : One to Many (1:M)

Sample output:-

- Customer (1) -- (M) Account
- Account (1) -- (M) Transaction
- Customer (1) -- (M) Loan
- Branch (1) -- (M) Employee
- Branch (1) -- (M) Account

Step (E): Reframing the Relations with key and constraints.

Convert the conceptual model into Relations (Tables)
with keys

Sample output

1. Customer (customer-ID [PK], Name, Address, phone, Email)
2. Branch (Branch-ID [PK], Branch-Name, location)
3. Account (Account-Number [PK], Account-Type, Balance, Date-opened, customer-ID [FK], Branch-ID [FK])
4. Transaction (Transaction-ID [PK], Transaction-Type, Amount, Date-time, Account-Number [FK])
5. Loan (Loan-ID [PK], Loan-Type, Loan-Amount, Interest-Rate, Start-date, customer-ID [FK])

6. Employee (Employee - ID [PK], Name, Position, Contact_No, SSN, Branch - ID [FK])

Constraints:

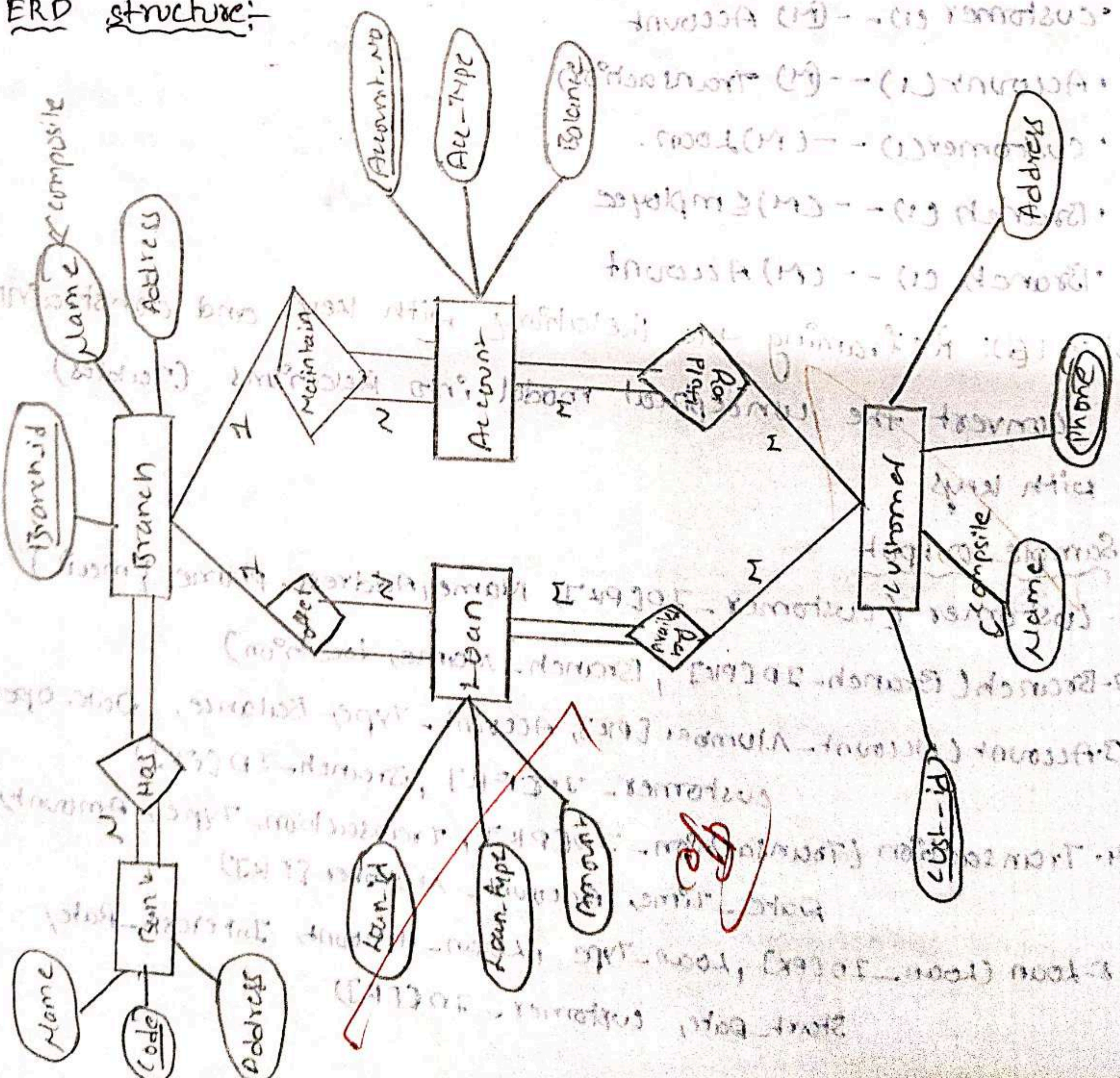
- PK: Unique identifiers for each table.
- FK: Foreign keys maintain referential integrity
- TM: enforced using foreign key relationships.

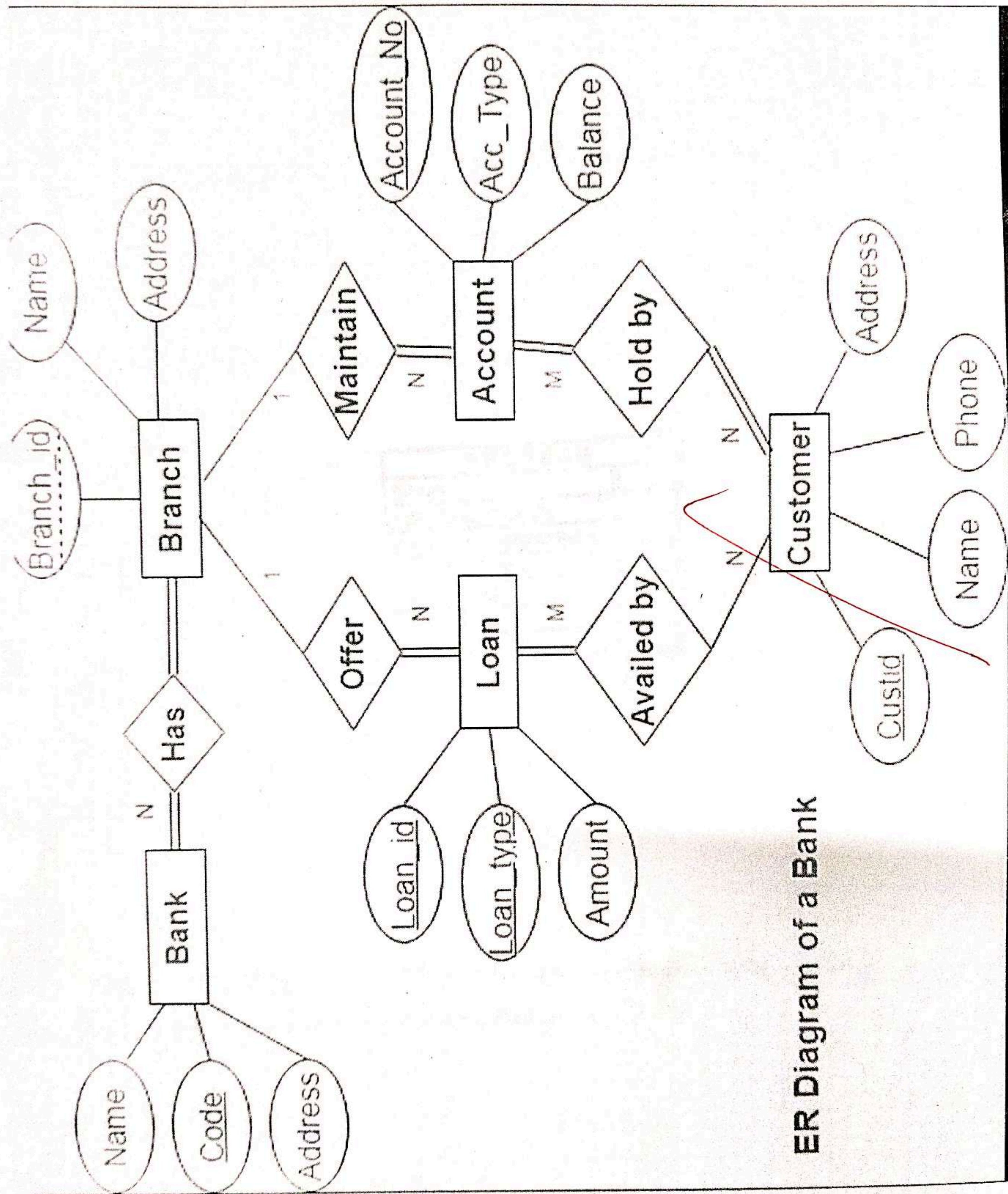
step(1c): Develop IER Diagram

By using draw.io to represent:

- Entities as rectangles
- Attributes as ovals
- Relationships as diamonds
- Indicate cardinalities (1:M)

ERD structure:





ER Diagram of a Bank